Logistic Regression

Logistic Regression is for **binary classification tasks**. It is mainly employed when the target variable is categorical with two possible classes or labels, typically represented as 0 and 1 or "Yes" and "No".

For example

1) Fraud or Not
2) Email spam or not Spam
3) Infected with Disease or not Disease.

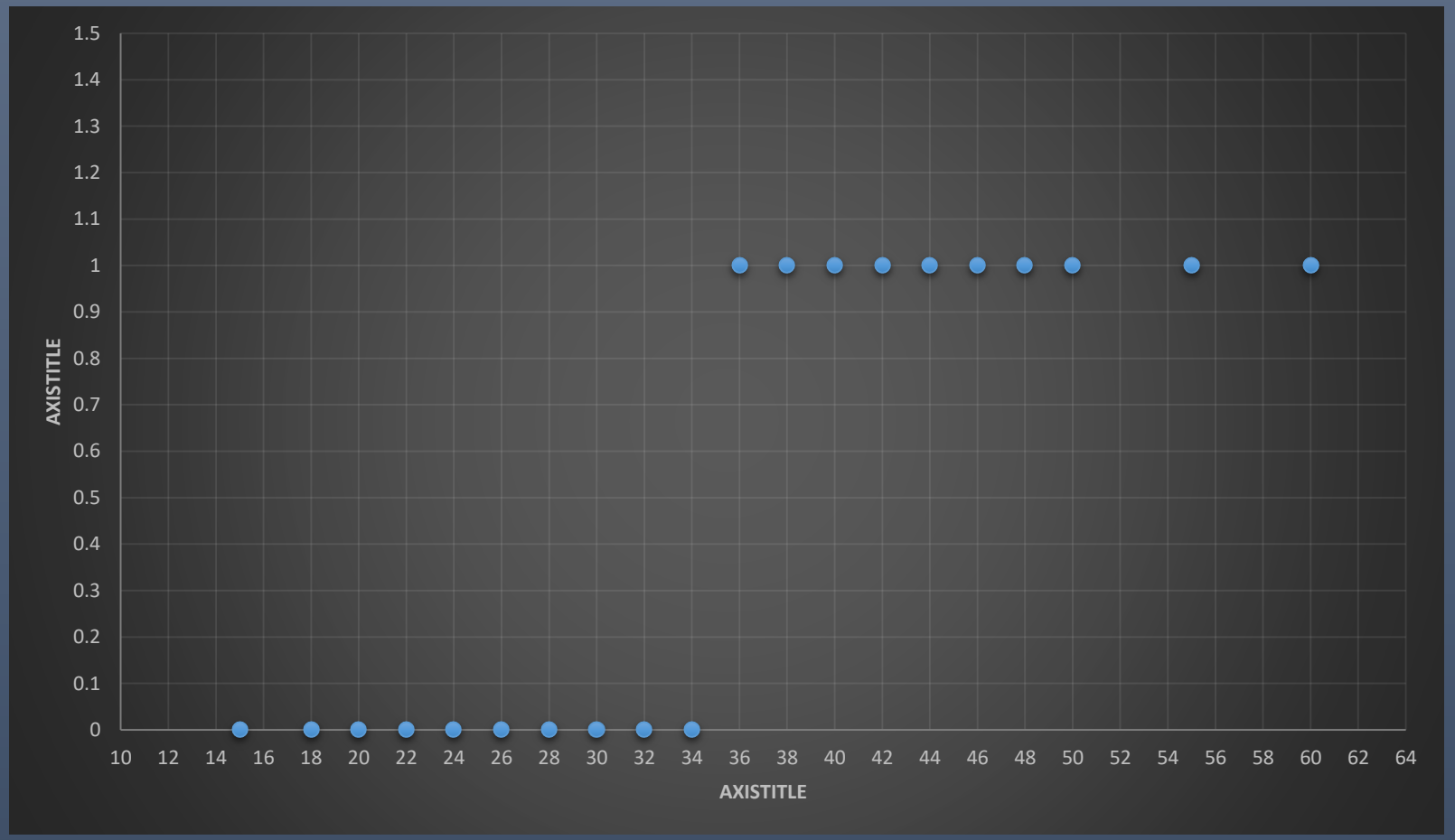
There are two types of approaches to compute logistic regression Graphical and Probabilistic approach.

Binary Class	Multi Class
from sklearn.linear_model import LogisticRegression model = LogisticRegression()	from sklearn.linear_model import LogisticRegression model = LogisticRegression(multi_class='ovr')

Initialize slope m and intercept c.
Calculate z = mx + c.
Apply the sigmoid function to get the predicted probability.
Calculate the error/loss.
Update m and c using gradient descent.
Repeat steps for multiple iterations.
Stop when the loss converges or the maximum iterations are reached.
Use the final m and c to make predictions.

Monthly Income (₹1000s)	Actual Loan Approved	mx+c	z	AfterSigmoid	0and1	Final Correct
15	0					
18	0					
20	0					
22	0					
24	0					
26	0					
28	0					
30	0					
32	0					
34	0					
36	1					
38	1					
40	1					
42	1					
44	1					
46	1					
48	1					
50	1					
55	1					
60	1					

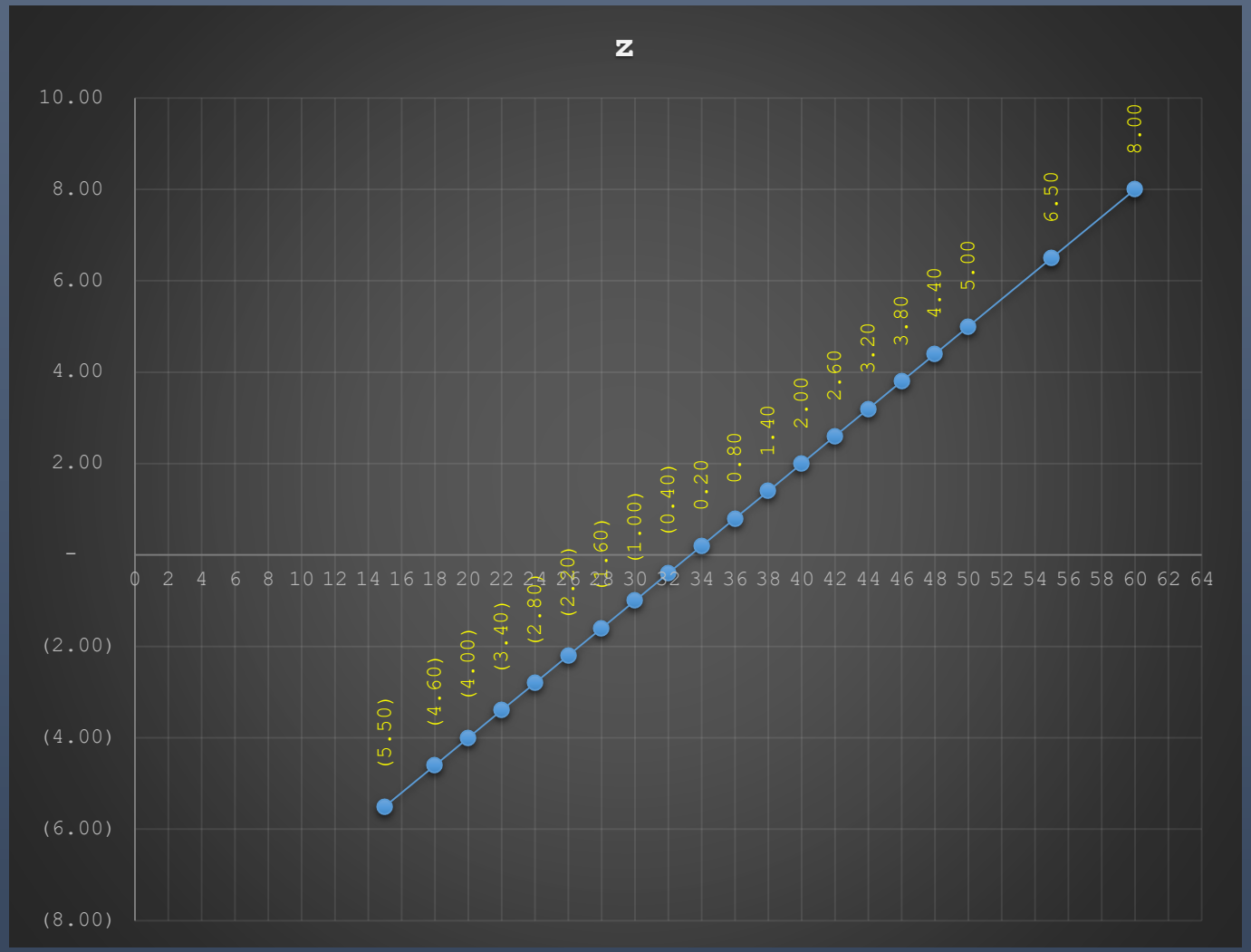
Monthly Income (₹1000s)	ActualLoanApproved
15	0
18	0
20	0
22	0
24	0
26	0
28	0
30	0
32	0
34	0
36	1
38	1
40	1
42	1
44	1
46	1
48	1
50	1
55	1
60	1



Intercept(c)= -10 Slope = 0.3

Monthly Income (₹1000s)	Actual Loan Approved	mx+c	z
15	0	=(0.3*15)+-10	-5.50
18	0	=(0.3*18)+-10	-4.60
20	0	=(0.3*20)+-10	-4.00
22	0	=(0.3*22)+-10	-3.40
24	0	=(0.3*24)+-10	-2.80
26	0	=(0.3*26)+-10	-2.20
28	0	=(0.3*28)+-10	-1.60
30	0	=(0.3*30)+-10	-1.00
32	0	=(0.3*32)+-10	-0.40
34	0	=(0.3*34)+-10	0.20
36	1	=(0.3*36)+-10	0.80
38	1	=(0.3*38)+-10	1.40
40	1	=(0.3*40)+-10	2.00
42	1	=(0.3*42)+-10	2.60
44	1	=(0.3*44)+-10	3.20
46	1	=(0.3*46)+-10	3.80
48	1	=(0.3*48)+-10	4.40
50	1	=(0.3*50)+-10	5.00
55	1	=(0.3*55)+-10	6.50
60	1	=(0.3*60)+-10	8.00

$$z = mx + c$$



Monthly Income (₹1000s)	Actual Loan Approved	mx+c	z	After Sigmoid
15	0	=(0.3*15)+-10	-5.50	0.00
18	0	=(0.3*18)+-10	-4.60	0.01
20	0	=(0.3*20)+-10	-4.00	0.02
22	0	=(0.3*22)+-10	-3.40	0.03
24	0	=(0.3*24)+-10	-2.80	0.06
26	0	=(0.3*26)+-10	-2.20	0.10
28	0	=(0.3*28)+-10	-1.60	0.17
30	0	=(0.3*30)+-10	-1.00	0.27
32	0	=(0.3*32)+-10	-0.40	0.40
34	0	=(0.3*34)+-10	0.20	0.55
36	1	=(0.3*36)+-10	0.80	0.69
38	1	=(0.3*38)+-10	1.40	0.80
40	1	=(0.3*40)+-10	2.00	0.88
42	1	=(0.3*42)+-10	2.60	0.93
44	1	=(0.3*44)+-10	3.20	0.96
46	1	=(0.3*46)+-10	3.80	0.98
48	1	=(0.3*48)+-10	4.40	0.99
50	1	=(0.3*50)+-10	5.00	0.99
55	1	=(0.3*55)+-10	6.50	1.00
60	1	=(0.3*60)+-10	8.00	1.00

$$SigmoidActivation = \frac{1}{1 + e^{-z}}$$

$$e = 2.718$$

$$= \frac{1}{1 + 2.718^{-(-5.50)}}$$

$$= 0.00$$

$$= \frac{1}{1 + 2.718^{-(-4.60)}}$$

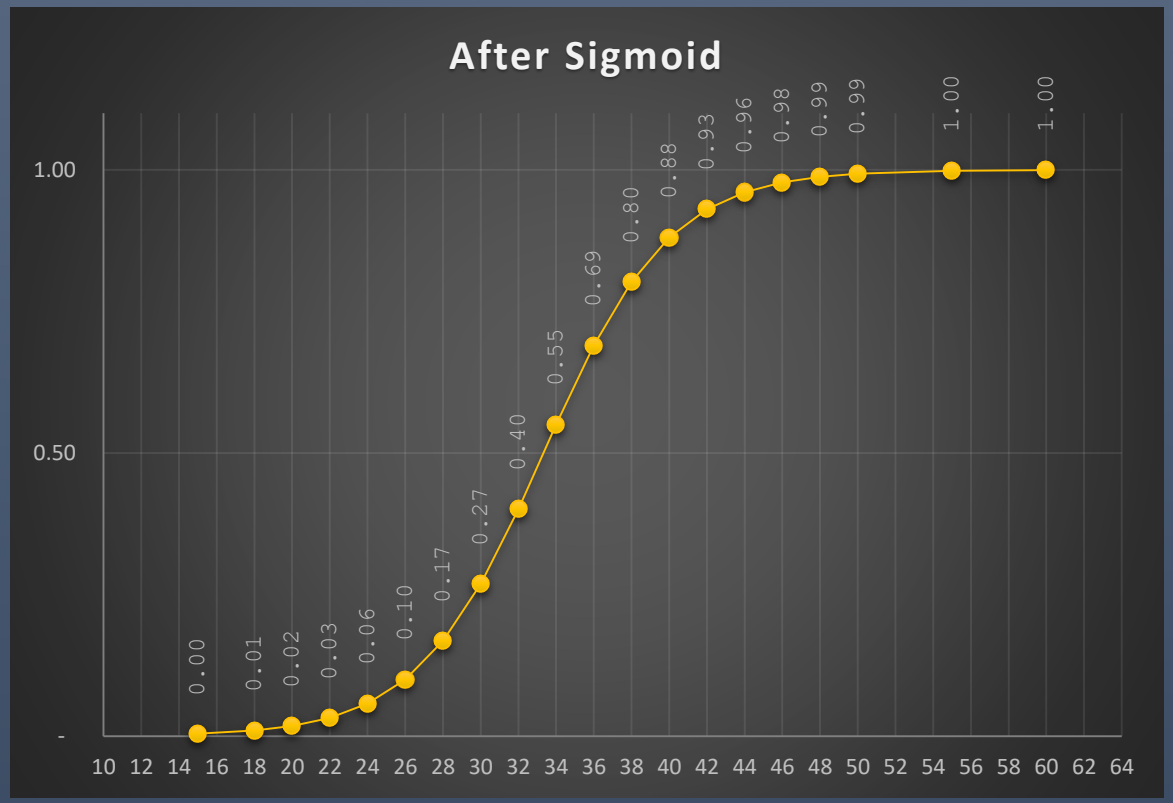
$$= 0.01$$

$$= \frac{1}{1 + 2.718^{-(-4.00)}}$$

$$= 0.02$$

Intercept(c)= -10 Slope = 0.3

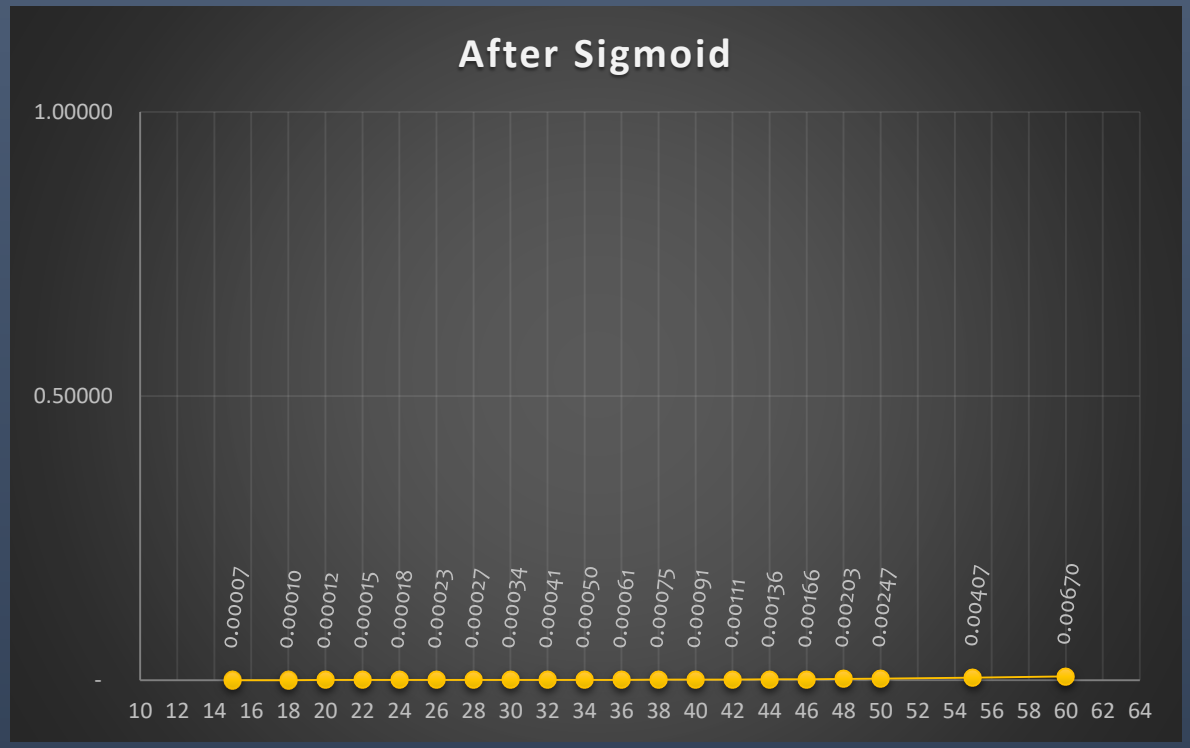
Monthly Income (₹1000s)	Actual Loan Approved	mx+c	z	After Sigmoid
15	0	=(0.3*15)+-10	-5.50	0.00
18	0	=(0.3*18)+-10	-4.60	0.01
20	0	=(0.3*20)+-10	-4.00	0.02
22	0	=(0.3*22)+-10	-3.40	0.03
24	0	=(0.3*24)+-10	-2.80	0.06
26	0	=(0.3*26)+-10	-2.20	0.10
28	0	=(0.3*28)+-10	-1.60	0.17
30	0	=(0.3*30)+-10	-1.00	0.27
32	0	=(0.3*32)+-10	-0.40	0.40
34	0	=(0.3*34)+-10	0.20	0.55
36	1	=(0.3*36)+-10	0.80	0.69
38	1	=(0.3*38)+-10	1.40	0.80
40	1	=(0.3*40)+-10	2.00	0.88
42	1	=(0.3*42)+-10	2.60	0.93
44	1	=(0.3*44)+-10	3.20	0.96
46	1	=(0.3*46)+-10	3.80	0.98
48	1	=(0.3*48)+-10	4.40	0.99
50	1	=(0.3*50)+-10	5.00	0.99
55	1	=(0.3*55)+-10	6.50	1.00
60	1	=(0.3*60)+-10	8.00	1.00



MonthlyIncome(₹1000s)	ActualLoanApproved	mx+c	mx+c	After Sigmoid	0and1	Final Correct
15	0	=(0.3*15)+-10	-5.50	0.00	0	Correct
18	0	=(0.3*18)+-10	-4.60	0.01	0	Correct
20	0	=(0.3*20)+-10	-4.00	0.02	0	Correct
22	0	=(0.3*22)+-10	-3.40	0.03	0	Correct
24	0	=(0.3*24)+-10	-2.80	0.06	0	Correct
26	0	=(0.3*26)+-10	-2.20	0.10	0	Correct
28	0	=(0.3*28)+-10	-1.60	0.17	0	Correct
30	0	=(0.3*30)+-10	-1.00	0.27	0	Correct
32	0	=(0.3*32)+-10	-0.40	0.40	0	Correct
34	0	=(0.3*34)+-10	0.20	0.55	1	-
36	1	=(0.3*36)+-10	0.80	0.69	1	Correct
38	1	=(0.3*38)+-10	1.40	0.80	1	Correct
40	1	=(0.3*40)+-10	2.00	0.88	1	Correct
42	1	=(0.3*42)+-10	2.60	0.93	1	Correct
44	1	=(0.3*44)+-10	3.20	0.96	1	Correct
46	1	=(0.3*46)+-10	3.80	0.98	1	Correct
48	1	=(0.3*48)+-10	4.40	0.99	1	Correct
50	1	=(0.3*50)+-10	5.00	0.99	1	Correct
55	1	=(0.3*55)+-10	6.50	1.00	1	Correct
60	1	=(0.3*60)+-10	8.00	1.00	1	Correct

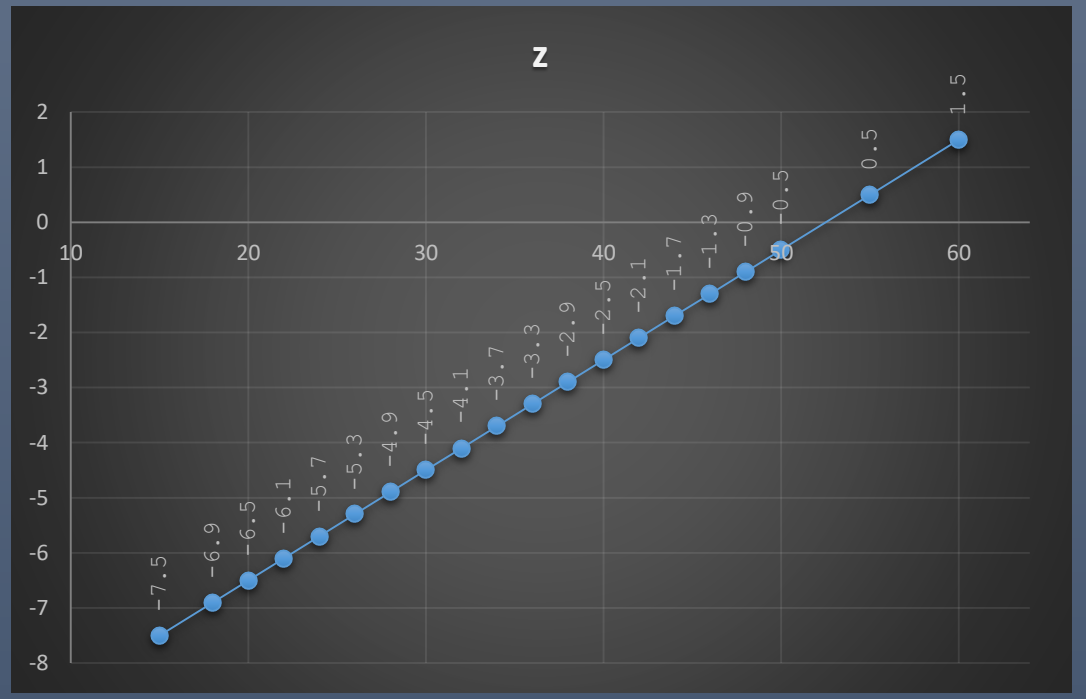
Intercept(c)= -11 Slope = 0.2

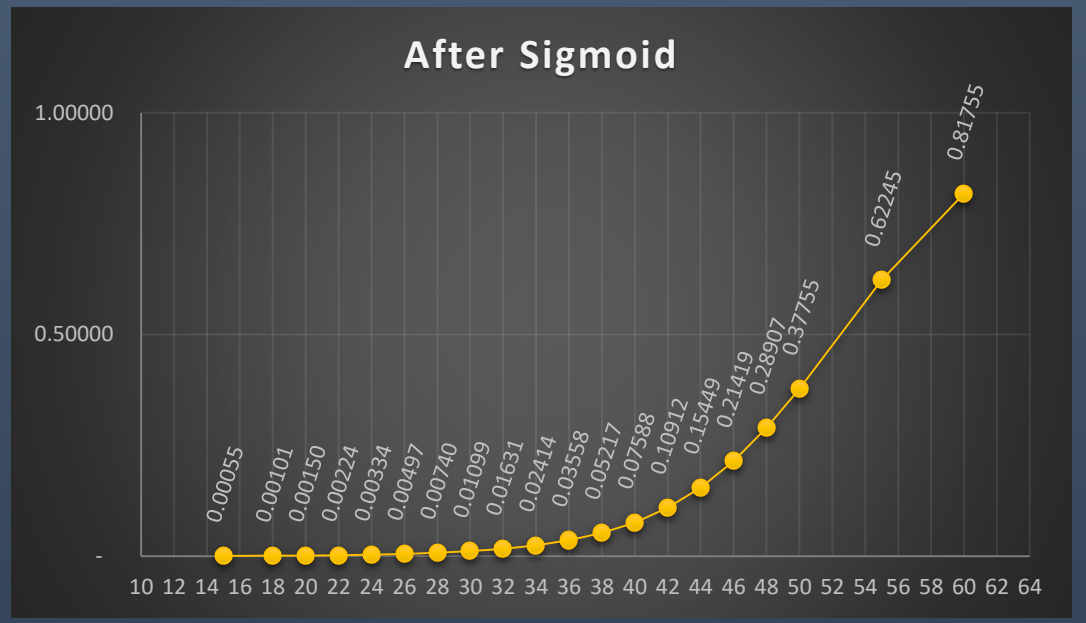
Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-9.5	0.00007	0	Correct
18	0	-9.2	0.00010	0	Correct
20	0	-9.0	0.00012	0	Correct
22	0	-8.8	0.00015	0	Correct
24	0	-8.6	0.00018	0	Correct
26	0	-8.4	0.00023	0	Correct
28	0	-8.2	0.00027	0	Correct
30	0	-8.0	0.00034	0	Correct
32	0	-7.8	0.00041	0	Correct
34	0	-7.6	0.00050	0	Correct
36	1	-7.4	0.00061	0	-
38	1	-7.2	0.00075	0	-
40	1	-7.0	0.00091	0	-
42	1	-6.8	0.00111	0	-
44	1	-6.6	0.00136	0	-
46	1	-6.4	0.00166	0	-
48	1	-6.2	0.00203	0	-
50	1	-6.0	0.00247	0	-
55	1	-5.5	0.00407	0	-
60	1	-5.0	0.00670	0	-



Intercept(c)= -10.5 Slope = 0.2

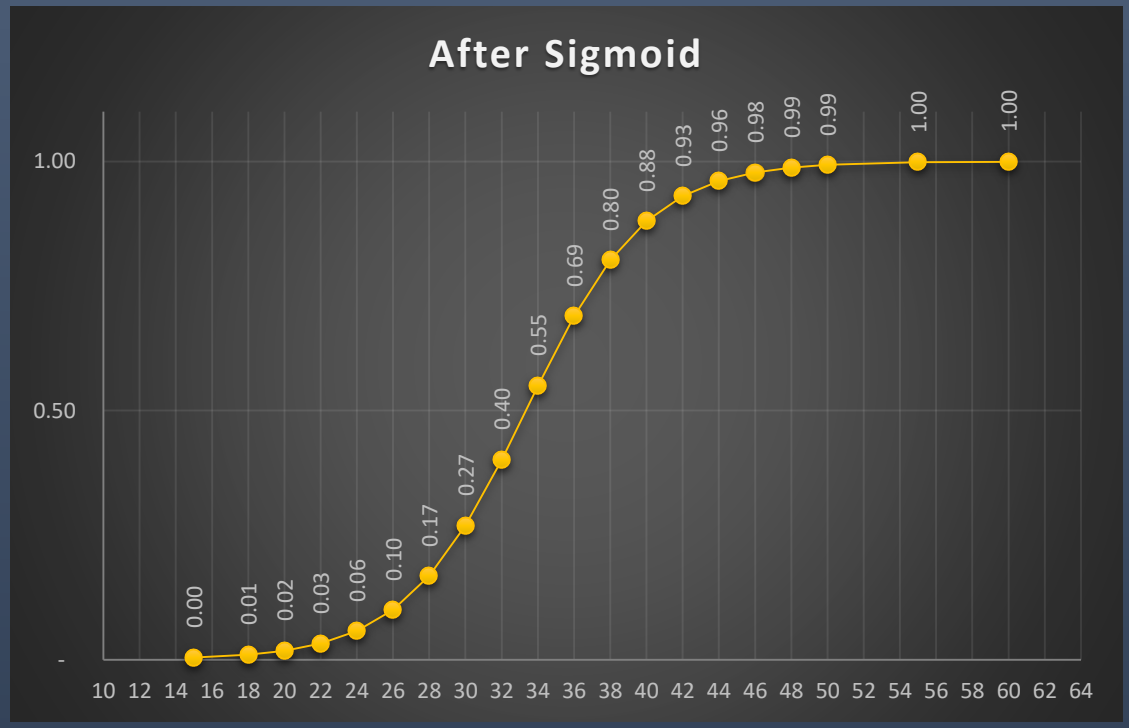
Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-7.5	0.00055	0	Correct
18	0	-6.9	0.00101	0	Correct
20	0	-6.5	0.00150	0	Correct
22	0	-6.1	0.00224	0	Correct
24	0	-5.7	0.00334	0	Correct
26	0	-5.3	0.00497	0	Correct
28	0	-4.9	0.00740	0	Correct
30	0	-4.5	0.01099	0	Correct
32	0	-4.1	0.01631	0	Correct
34	0	-3.7	0.02414	0	Correct
36	1	-3.3	0.03558	0	-
38	1	-2.9	0.05217	0	-
40	1	-2.5	0.07588	0	-
42	1	-2.1	0.10912	0	-
44	1	-1.7	0.15449	0	-
46	1	-1.3	0.21419	0	-
48	1	-0.9	0.28907	0	-
50	1	-0.5	0.37755	0	-
55	1	0.5	0.62245	1	Correct
60	1	1.5	0.81755	1	Correct





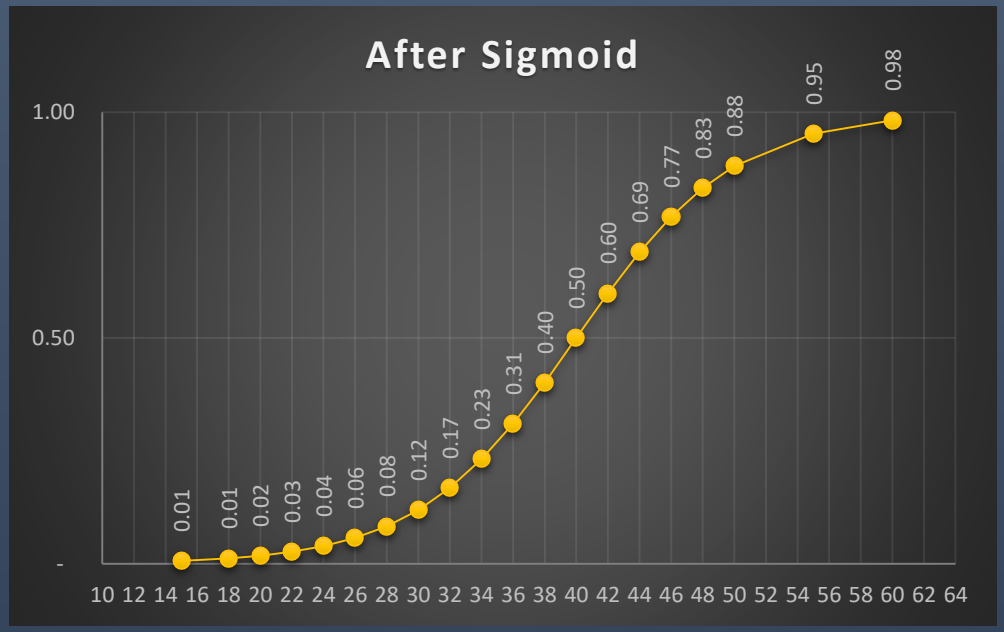
Intercept(c)= -10 Slope = 0.3

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-5.50	0.00	0	Correct
18	0	-4.60	0.01	0	Correct
20	0	-4.00	0.02	0	Correct
22	0	-3.40	0.03	0	Correct
24	0	-2.80	0.06	0	Correct
26	0	-2.20	0.10	0	Correct
28	0	-1.60	0.17	0	Correct
30	0	-1.00	0.27	0	Correct
32	0	-0.40	0.40	0	Correct
34	0	0.20	0.55	1	-
36	1	0.80	0.69	1	Correct
38	1	1.40	0.80	1	Correct
40	1	2.00	0.88	1	Correct
42	1	2.60	0.93	1	Correct
44	1	3.20	0.96	1	Correct
46	1	3.80	0.98	1	Correct
48	1	4.40	0.99	1	Correct
50	1	5.00	0.99	1	Correct
55	1	6.50	1.00	1	Correct
60	1	8.00	1.00	1	Correct



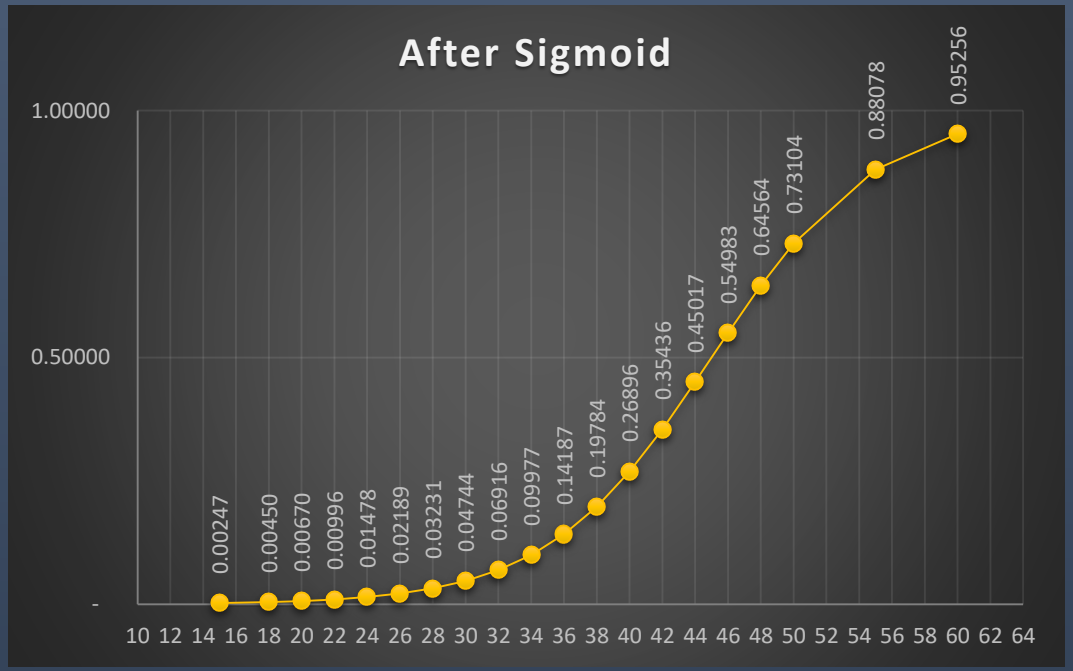
Intercept(c)= -8 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0and1	Final Correct
15	0	-5.00	0.01	0	Correct
18	0	-4.40	0.01	0	Correct
20	0	-4.00	0.02	0	Correct
22	0	-3.60	0.03	0	Correct
24	0	-3.20	0.04	0	Correct
26	0	-2.80	0.06	0	Correct
28	0	-2.40	0.08	0	Correct
30	0	-2.00	0.12	0	Correct
32	0	-1.60	0.17	0	Correct
34	0	-1.20	0.23	0	Correct
36	1	-0.80	0.31	0	-
38	1	-0.40	0.40	0	-
40	1	-	0.50	1	Correct
42	1	0.40	0.60	1	Correct
44	1	0.80	0.69	1	Correct
46	1	1.20	0.77	1	Correct
48	1	1.60	0.83	1	Correct
50	1	2.00	0.88	1	Correct
55	1	3.00	0.95	1	Correct
60	1	4.00	0.98	1	Correct

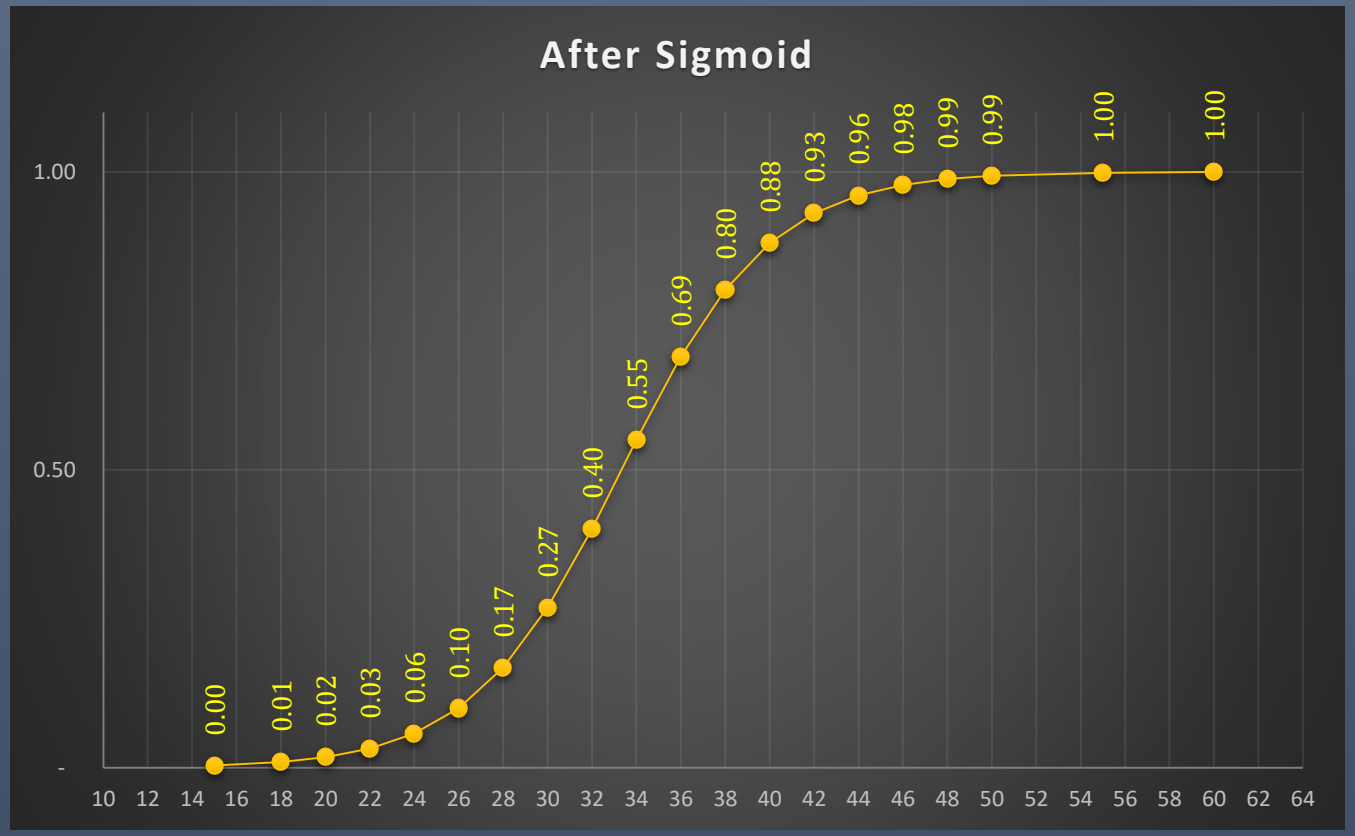


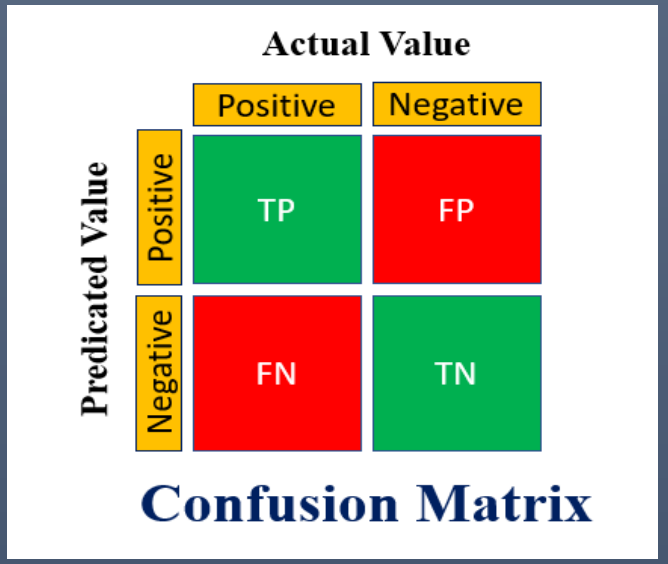
Intercept(c)= -9 Slope = 0.2

Monthly Income (₹1000s)	Actual Loan Approved	z	After Sigmoid	0 and 1	Final Correct
15	0	-6.00	0.00247	0	Correct
18	0	-5.40	0.00450	0	Correct
20	0	-5.00	0.00670	0	Correct
22	0	-4.60	0.00996	0	Correct
24	0	-4.20	0.01478	0	Correct
26	0	-3.80	0.02189	0	Correct
28	0	-3.40	0.03231	0	Correct
30	0	-3.00	0.04744	0	Correct
32	0	-2.60	0.06916	0	Correct
34	0	-2.20	0.09977	0	Correct
36	1	-1.80	0.14187	0	-
38	1	-1.40	0.19784	0	-
40	1	-1.00	0.26896	0	-
42	1	-0.60	0.35436	0	-
44	1	-0.20	0.45017	0	-
46	1	0.20	0.54983	1	Correct
48	1	0.60	0.64564	1	Correct
50	1	1.00	0.73104	1	Correct
55	1	2.00	0.88078	1	Correct
60	1	3.00	0.95256	1	Correct



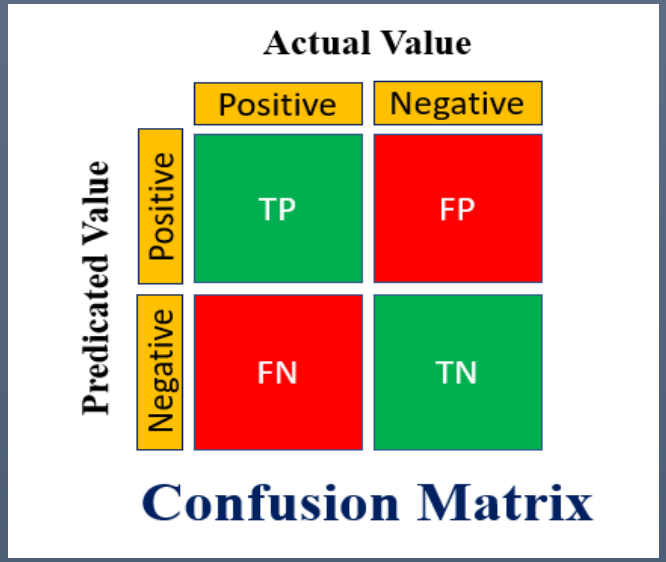
Intercept	slope	Correct	Incorrect
-11.00	0.10	10	10
-10.50	0.20	12	8
-10.00	0.30	19	1
-8.00	0.20	18	2
-9.00	0.20	15	5




Actual Value
Positive
Negative
Predicated Value
Positive
TP
FP
Negative
FN
TN
Confusion Matrix

Evaluation Metrix for classification(Confusion matrix)

1) Accuracy
2) Precision
3) Recall
4) F-Beta Square

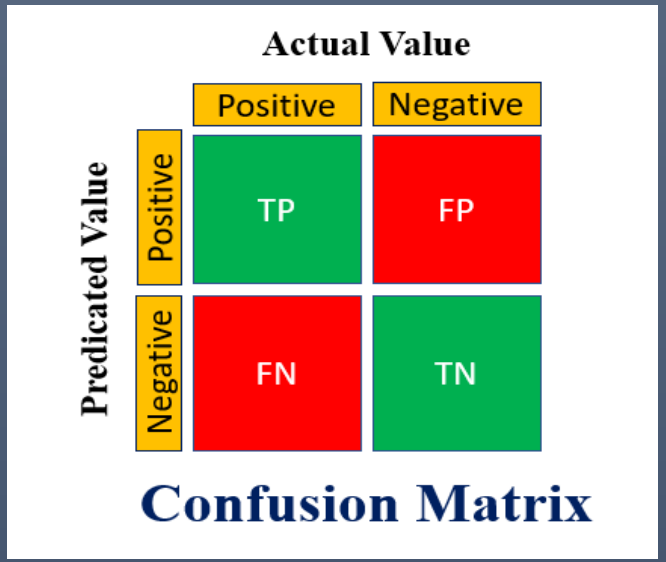


Confusion matrix :

The confusion matrix can be explained by studying the graph below.

Accuracy:

The accuracy function will predict the overall performance of the model.



$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

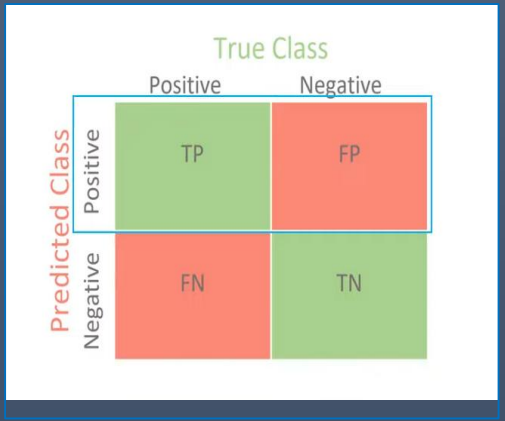
$$Accuracy = \frac{\text{Total Correctly Predicted Value}}{\text{Total Value}}$$

Precision

We focus on Precision when It is important to evaluate and **minimize "false positives"**, especially in sensitive domains where false negatives are less of a concern compared to false positives.

High Precision means that the model is effective where we want to minimize **"false positives"**.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \Rightarrow \text{Precision} = \frac{\text{True Positive}}{\text{Total Predicted Positive}}$$



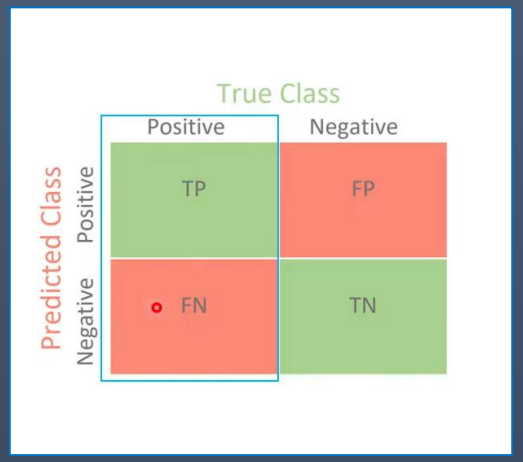
Example :

Face recognition:

Recall:

The recall is particularly useful in situations where it is important to **minimize "false negatives"**. For example, in medical diagnostics, it is crucial to identify as many true positive cases as possible, even if it results in a higher number of false positives. In such cases, recall serves as a valuable metric to evaluate the model's ability to identify positive instances.

High recall means that the model is effective when we want to minimize **"false negatives"**.



$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \Rightarrow \text{Recall} = \frac{\text{True Positive}}{\text{Total Actual Positive}}$$

F-1measure:

If both True positive and False Negative is important.

Ie precision and recall are important

$$F_\beta = (1 + \beta^2) \cdot \frac{precision \cdot recall}{(\beta^2 \cdot precision) + recall}$$